HW #13.

1. From the figure.

(a) $X(z)+b_1z^{-1}Y_1(z)=Y_1(z) \Rightarrow H_1(z)=\frac{Y_1(z)}{X(z)}=\frac{1}{1-b_1z^{-1}}$

$X(z)+b_2z^{-1}Y_2(z)=Y_2(z) \Rightarrow H_2(z)=\frac{Y_2(z)}{X(z)}=\frac{1}{1-b_2z^{-1}}$

So that $H(z)=H_1(z)+H_2(z)=\frac{1}{1-b_1z^{-1}}+\frac{1}{1-b_2z^{-1}}$

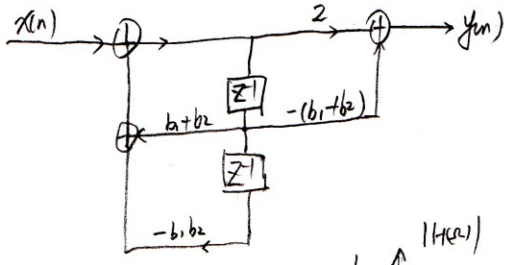
(b) $H(z)=\frac{Y(z)}{X(z)}=\frac{1}{1-b_1z^{-1}}+\frac{1}{1-b_2z^{-1}}$

$\frac{Y(z)}{X(z)}=\frac{1-b_2z^{-1}+1-b_1z^{-1}}{(1-b_1z^{-1})(1-b_2z^{-1})}=\frac{2-(b_1+b_2)z^{-1}}{1-(b_1+b_2)z^{-1}+b_1b_2z^{-2}}$

$Y(z)-(b_1+b_2)z^{-1}Y(z)+b_1b_2z^{-2}Y(z)=2X(z)-(b_1+b_2)z^{-1}X(z)$

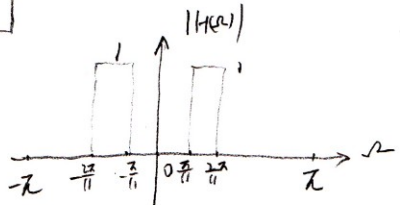
$\Rightarrow y(n)-(b_1+b_2)y(n-1)+b_1b_2y(n-2)=2x(n)-(b_1+b_2)x(n-1)$

(c) Direct-form II.



2. (a)

step 1: $\Omega_c=\frac{\omega_c}{F_s}=\frac{2\pi f_c}{F_s}$



step 2: $H_d(\Omega)=\begin{cases}1, & -\frac{2\pi}{11}\le\Omega\le\frac{\pi}{11} \text{ and } \frac{\pi}{11}\le\Omega\le\frac{2\pi}{11}\\ 0, & 0.W\end{cases}$

$h_d(n)=\frac{1}{2\pi}\int_{-\pi}^{\pi}H_d(\Omega)e^{j\Omega n}d\Omega=\frac{1}{2\pi}\cdot\int_{-\frac{2\pi}{11}}^{\frac{\pi}{11}}e^{j\Omega n}d\Omega+\frac{1}{2\pi}\int_{\frac{\pi}{11}}^{\frac{2\pi}{11}}e^{j\Omega n}d\Omega$

$=\frac{1}{2\pi jn}\left[e^{-j\frac{\pi}{11}n}-e^{-j\frac{2\pi}{11}n}+e^{j\frac{2\pi}{11}n}-e^{j\frac{\pi}{11}n}\right]$

step 3: $h(n)=h_d[n-5]\cdot w(n)$ $=\frac{1}{2\pi jn}\left[2j\sin\frac{2\pi}{11}n-2j\sin\frac{\pi}{11}n\right]$

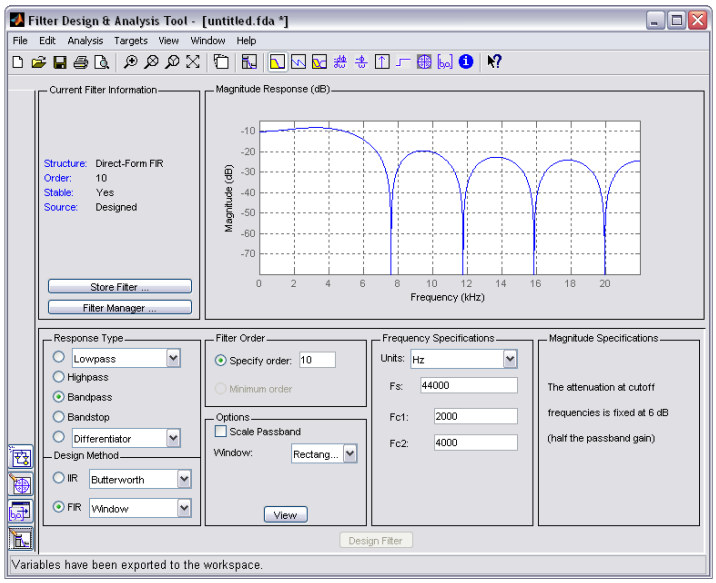
$=\frac{1}{\pi n}\left[\sin\frac{2\pi}{11}n-\sin\frac{\pi}{11}n\right]$

$=\frac{2}{11}\mathrm{sinc}\left(\frac{2}{11}n\right)-\frac{1}{11}\mathrm{sinc}\left(\frac{1}{11}n\right)$

```
N=11;
n=[-(N-1)/2:(N-1)/2];
hd_n=2*sinc(2*n/11)/11-sinc(n/11)/11;
wd=window(@rectwin,N);
hn=hd_n.*wd.';
```

hn=[-0.0451 -0.0122 0.0248 0.0587 0.0824 0.0909 0.0824 0.0587 0.0248 -0.0122 -0.0451]

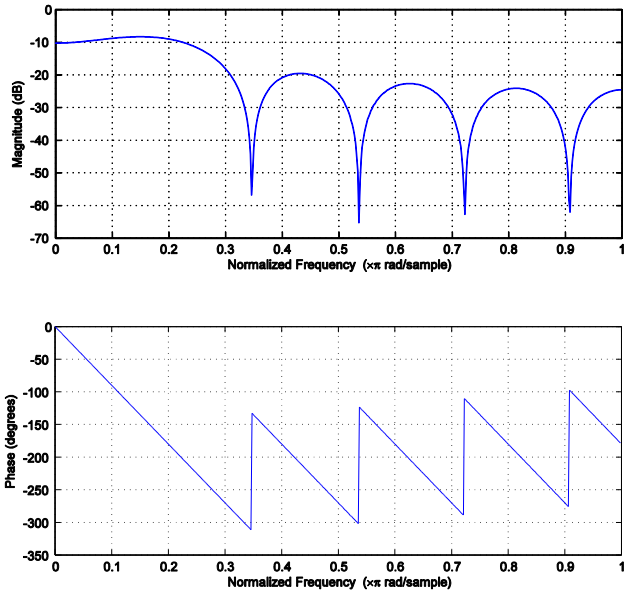
Confirm with fdatool from matlab.



Coefficient

Num =[-0.0451 -0.0122 0.0248 0.0587 0.0824 0.0909 0.0824 0.0587 0.0248 -0.0122 -0.0451]

The amplitude and phase response



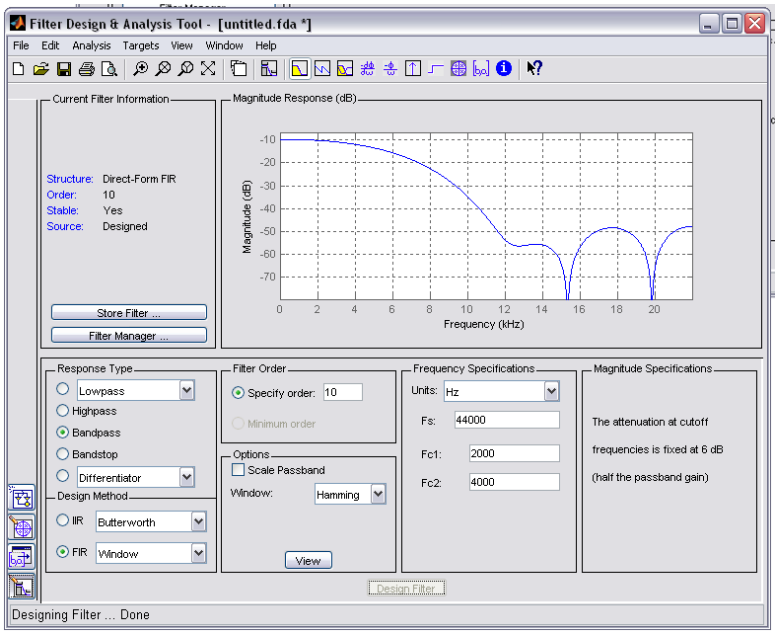
(b) If the Hamming window is used

```
N=11;
n=[-(N-1)/2:(N-1)/2];
hd_n=2*sinc(2*n/11)/11-sinc(n/11)/11;
wd=window(@hamming,N);
hn=hd_n.*wd.';
```

hn =[-0.0036 -0.0021 0.0099 0.0401 0.0752 0.0909 0.0752 0.0401 0.0099

-.0021 -0.0036]

Confirm with fdatool from matlab

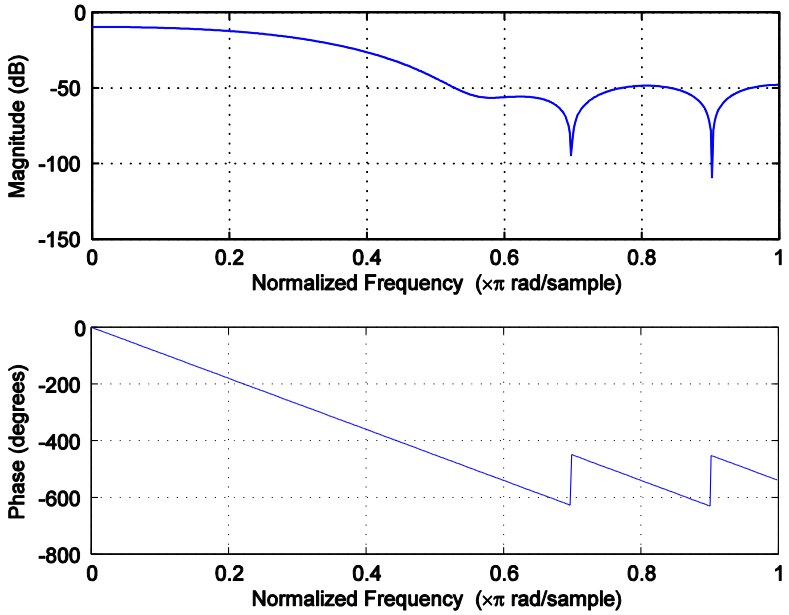


Coefficients:

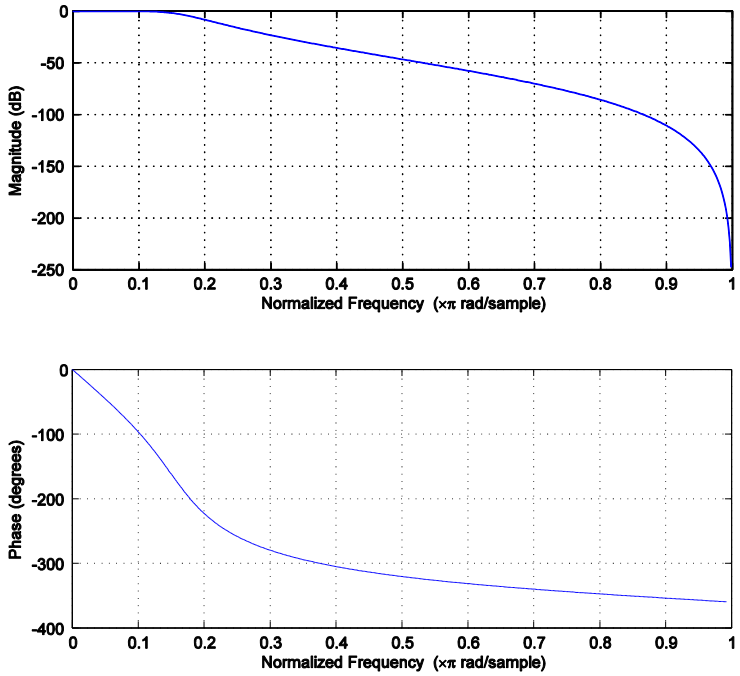
Num =[-0.0036 -0.0021 0.0099 0.0401 0.0752 0.0909 0.0752 0.0401 0.0099

-.0021 -0.0036]

The amplitude and phase response



3.




```
% analog Butterworth filter, in the form of zeros and
poles
% cut-off frequency 1 rad/s
[z,p,k] = buttap(4); % creates a 2-pole Butterworth
filter
% transfer function
[num,den] = zp2tf(z,p,k); % poles & zeros to transfer
function
% need to transform the cutoff frequency to 2k Hz
fc=2000;
wc = 2*pi*fc;
[num,den] = lp2lp(num,den,wc);
% sampling frequency 24k Hz
Fs = 24000;
% generate the digital filter with the bilinear
transformation
[numd,dend] = bilinear(num,den,Fs);
freqz(numd, dend)
```